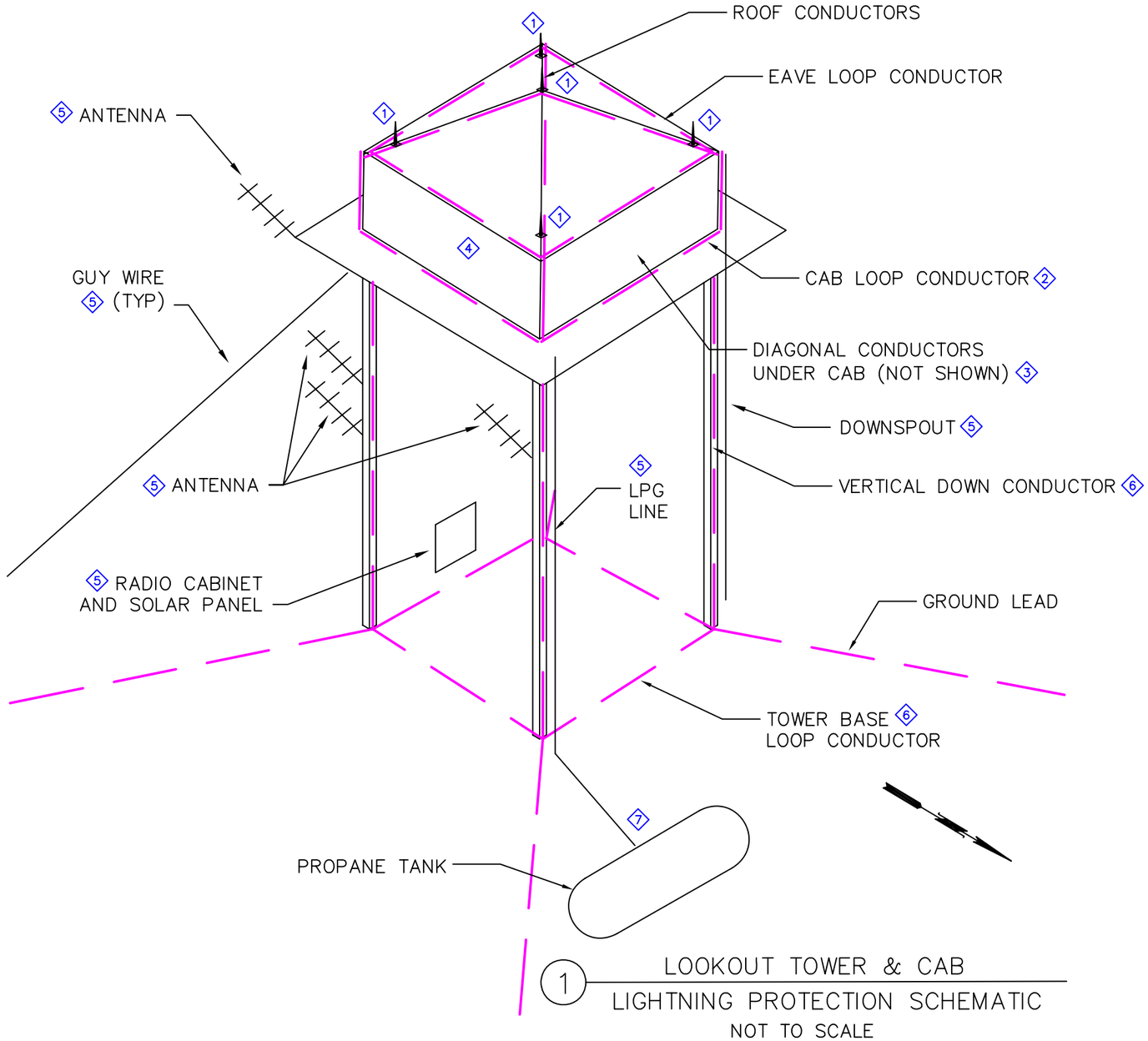
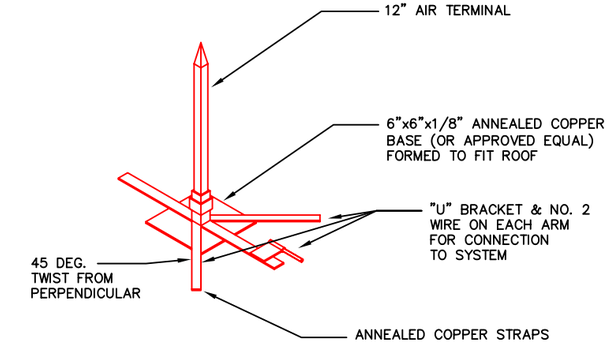
USFS Seed2 v1.0



1 LOOKOUT TOWER & CAB
LIGHTNING PROTECTION SCHEMATIC
NOT TO SCALE



2 LIGHTNING PROTECTION
12" AIR TERMINAL DETAIL
NOT TO SCALE

GENERAL NOTES:

1. THIS SCHEMATIC IS TO BE USED TO IDENTIFY THE LIGHTNING PROTECTION WORK ITEMS AND TO SHOW THE GENERAL CAB AND TOWER LIGHTNING PROTECTION LAYOUT FOR REFERENCE ONLY. THIS SCHEMATIC IS NOT TO SCALE AND DOES NOT INCLUDE ALL PERTINENT INFORMATION. IT IS RECOMMENDED THAT THE CONTRACTOR VERIFY THE TYPE AND QUANTITY OF ITEMS NECESSARY FOR THIS WORK PRIOR TO STARTING WORK.
2. ALL LIGHTNING PROTECTION WORK SHALL BE INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF THE SPECIFICATIONS AND THE MOST CURRENT VERSION OF THE NFPA 78 LIGHTNING PROTECTION CODE. IF THERE IS A DISCREPANCY BETWEEN THESE DRAWINGS OR SPECIFICATIONS AND THE NFPA 78 CODE, THE CODE SHALL GOVERN.
3. ALL CONDUCTORS SHALL BE #2/0 SOFT DRAWN SOLID COPPER WIRE OR APPROVED EQUAL.
4. ALL CONDUCTORS SHALL BE PROPERLY FASTENED EVERY 3'–0" WITH APPROPRIATE STRAPS OR FASTENERS.
5. ALL CONDUCTOR BENDS SHALL HAVE A MINIMUM RADIUS OF 8".
6. ALL CONDUCTORS SHALL BE CONTINUOUS WHERE STATED.

SPECIFIC WORK ITEMS:

1. INSTALL FIVE (5) NEW AIR TERMINALS ON THE ROOF. EACH OF THESE AIR TERMINALS SHALL BE PLACED APPROXIMATELY 12" IN FROM THE EDGE OF THE ROOF. ONE WILL BE INSTALLED AT THE PEAK. EACH NEW AIR TERMINAL SHALL BE PROPERLY FASTENED TO THE ROOF. INSTALL NEW EAVE LOOP CONDUCTOR FOR THE CONNECTION TO THE PROTECTION SYSTEM.
2. REPLACE THE UNDERFLOOR CAB LOOP CONDUCTOR AND CONNECT TO THE PROTECTION SYSTEM.
3. INSTALL A NEW DIAGONAL CONDUCTOR UNDERNEATH THE CAB FLOOR. CONNECT THE DIAGONAL TO THE PROTECTION SYSTEM.
4. PROPERLY TIE ALL METAL OBJECTS INSIDE THE CAB TO THE UNDERFLOOR CAB CONDUCTOR LOOP. PROPERLY SEAL ALL WALL AND FLOOR PENETRATIONS.
5. PROPERLY TIE ANY METAL OBJECTS OUTSIDE OF THE LOOKOUT CAB ON THE CATWALK TO THE NEW CAB LOOP CONDUCTOR UNDER THE CATWALK:
   - RADIO ANTENNAE, RADIO CABINET AND RADIO SOLAR CHARGING SYSTEM
   - DOWNSPOUT
   - TOP OF EACH GUY WIRE
   - PROPANE LINE ON THE TOWER STRUCTURE
6. CONNECT ALL VERTICAL CONDUCTORS TO EXISTING IN–GROUND LOOP.
7. PROPERLY TIE OBJECTS NEAR THE BASE OF THE TOWER TO THE EXISTING IN–GROUND LOOP:
   - PROPANE LINE AT BOTTOM OF TOWER
   - PROPANE TANK
8. PROPERLY SECURE ALL NEW CONDUCTORS AT 3' O.C. THIS INCLUDES ALL ROOF CONDUCTORS, THE ROOF EAVE LOOP, THE CAB LOOP, THE DIAGONAL UNDER THE CAB, THE TOWER BASE LOOP, AND THE FOUR VERTICAL DOWN CONDUCTORS ON THE TOWER LEGS.

U.S. DEPARTMENT OF AGRICULTURE
FOREST SERVICE
R–3
SOUTHWESTERN REGION

Drawn	T.WILLIAMS
Design	S.HODGES
Checked	S.CLARK
Reviewed	J.BOOTH

Forest
COCONINO NATIONAL FOREST

Project Name
EAST POCKET LOOKOUT
REHABILITATION

Sheet Title
LIGHTNING PROTECTION
SCHEMATIC AND DETAILS

Scale
N.T.S.

Sheet
5
of 5